



BEFORE

In 1861, when the Civil War began, ironclad warships were novelties in naval ship design. Not surprisingly, both the Union and the Confederates sought to employ this new technology.

During the Crimean War, the French and British navies **attached iron plates** to the **wooden hulls** of some of their warships as armor. The experiment was considered a success and **in 1859 France launched the *Gloire***, an ironclad battleship. The British responded with their own armored battleship, *Warrior*, the following year.

DEVELOPMENTS IN THE UNITED STATES

The Confederacy saw the possibility of using an ironclad to challenge Union naval supremacy. In the first days of the war they **captured Norfolk Navy Yard** and with it the burned-out hull of the steam-and-sail frigate **USS *Merrimack***. The ship was salvaged and **rebuilt as the steam-driven "ironclad" CSS *Virginia***, encased to the waterline in **4in (10cm) iron plate armor**. Meanwhile, the Union raced to produce the far **more radical all-metal USS *Monitor*** in time to confront the *Virginia* on her first sortie.

10 The guns carried by CSS *Virginia*—two 7-in rifled guns, fore and aft, six 9-in smoothbore Dahlgren guns in broadside, and two 6.4-in rifled guns.

The Confederacy was desperate to break the Union naval blockade that cut off Richmond's access to international trade through the James River. On March 8, 1862, the ironclad CSS *Virginia* steamed out of the Norfolk Navy Yard to challenge the Union blockade force in Hampton Roads, where the James River opens out into Chesapeake Bay. To the crews of the Union warships on blockade duty, the ungainly, slow-moving warship with her sloped ironplate armor was a sinister sight. The *Virginia's* captain, 61-year-old Franklin Buchanan, had an array of guns at his disposal, but his intended principal weapon was an iron ram fitted to his ship's prow. As he steamed toward the sloop USS *Cumberland*, the Union ship's shells bounced off *Virginia's* metal plates. The ram tore into the *Cumberland's* wooden hull and sent her to the bottom—almost taking the *Virginia* with her as the ram stayed locked in the sloop's hull. Fortunately for Buchanan and his crew, the ram snapped off and the *Virginia* was able to turn her attention to the frigate USS *Congress*. Having witnessed the *Cumberland's* fate, the captain of the *Congress* ran his ship aground to make it impossible for the Confederate ironclad to ram. This left the frigate a sitting target for the *Virginia's* guns and other vessels of the Confederate squadron.